Akoizo Original Practice - Form 1

Division C | Electric Vehicle | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A bottle pusher's widest opening is 12.0 cm. What is its Pusher Bonus?

A. -34.5

B. -23.0

C. +34.5

D. -18.0

Answer: _____

2. (2 points) What is the maximum vehicle width?

Unit: Power, construction limits, and legal parts.
Worked-case data: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

A. 30.0 cm

B. 50.0 cm

C. 35.0 cm

D. 80.0 cm

Answer: _____

3. (2 points) When does Run Time end?

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

A. When the MP first passes the Target Point

B. Exactly at the Target Time

C. When the vehicle has been stationary for 3 seconds (the 3 s not counted)

D. When the bottle crosses the line

Answer: _____

4. (2 points) Why must the pusher travel beyond the Bottle Line by more than zero?

A. The MP must cross the line

B. To reduce Run Time

C. The bottle’s near side must cross the line, so the pusher must go one bottle diameter past it

D. The rules require 10 cm past

Answer: _____

5. (2 points) What is the first tiebreaker?

Unit: Competition day: impound, check-in, and running.
Worked-case data: With 1:30 left in event time, a team’s first run was successful. Its vehicle now has a loose wire. What should it do?
Separate investigation observations:
Run 1: Vehicle started while the supervisor was still positioning timers.
Supervisor ruling: Run 1 recorded as a Failed Run.
Run 2: Clean run scored 96.4.

A. Better Distance Score of the scored run

B. Lower Time Score

C. Narrower pusher

D. Lighter vehicle

Answer: _____

6. (4 points) Why might a team choose a wider pusher than the narrowest possible?

Answer: _____
_____
_____

7. (4 points) Give two causes of wheel slip and one way to reduce it.

Unit: Drivetrain and distance sensing: motors, gears, and encoders.
Worked-case data: A 6.5 cm wheel drives a 100-count-per-revolution encoder through a 3:1 motor-to-wheel gear ratio, with the encoder on the motor shaft. How many counts for 8.60 m?
Separate investigation observations:
Forward test: Commanded 500 cm, measured 503.5 cm.
Reverse test: Commanded 150 cm back, measured 148.9 cm.
Tires: Rubber O-rings, slightly compressed under load.

Answer: _____
_____
_____

8. (4 points) Rank these by typical size of effect: Time Score, Pusher Bonus, construction violation.

Unit: Scoring a run: distance, time, bottle, and pusher.
Worked-case data: Target Time 14 s. Pusher opening 16.0 cm. The vehicle stops 22.5 cm from the Target Point in 16.37 s; the bottle ends 12.1 cm beyond the Bottle Line. Find the Run Score.
Separate investigation observations:
Pusher: Widest opening 20.0 cm.
Run 1: MP 6.0 cm off, bottle 5.0 cm past the line, time error 3.0 s.
Run 2: MP 3.0 cm off, bottle 8.0 cm short of the line, time error 0.5 s.

Answer: _____
_____
_____

9. (4 points) What two things must a team using a kit provide?

Unit: Power, construction limits, and legal parts.
Worked-case data: A team's vehicle uses six NiMH AA cells, a motor driver chip, a microcontroller, a 1 F supercapacitor to boost acceleration, and a laser pointer for aiming. What must change?
Separate investigation observations:
Parts list: 8 x AA alkaline, Arduino, encoder, 12 V boost converter.
Inspection question: Does the boost converter store energy to exceed battery power?
Team explanation: It only steps up voltage continuously from the batteries; no large storage.

Answer: _____
_____
_____

10. (4 points) What must the team not do once a run starts?

Unit: Control: speed profiles, stopping, and hitting the Target Time.
Worked-case data: Total MP travel is 8.60 m forward plus 1.60 m back. Target Time is 15.0 s. You allow 1.0 s for each of four speed ramps, 0.5 s to reverse, and 0.3 s of settling. What cruise speed is needed?
Separate investigation observations:
Fresh batteries: Fixed-power run: 13.4 s, stop error 4 cm.
Half-drained: Same code: 15.9 s, stop error 9 cm.
Closed-loop version: Both battery states: 15.0 ± 0.2 s, stop error 1-2 cm.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: Target Distance 7.00 m, Bottle Line Distance 2.00 m, vehicle length 50 cm, bottle 6.5 cm, desired margin 3 cm. How far does the MP travel forward and backward?

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: With 1:30 left in event time, a team's first run was successful. Its vehicle now has a loose wire. What should it do?

Answer: _____
_____
_____

13. (2 points) Which pusher should the team use?

Supplied synthetic observations:
Lateral error at bottle: Up to ±2.0 cm in 20 runs.
Pusher A: 8.0 cm opening, no funnel: 14 of 20 captures.
Pusher B: 11.0 cm opening with V entry: 20 of 20 captures.

A. Pusher B

B. Pusher A

C. No pusher

Answer: _____

14. (2 points) Define "Circumference" in the context of Electric Vehicle.

Answer: _____
_____
_____

15. (2 points) Define "Vehicle Distance" in the context of Electric Vehicle.

Answer: _____
_____
_____

16. (2 points) Define "Energy storage device" in the context of Electric Vehicle.

Answer: _____
_____
_____

17. (2 points) Define "PID controller" in the context of Electric Vehicle.

Answer: _____
_____
_____

18. (2 points) Define “Vehicle length offset” in the context of Electric Vehicle.

Answer: _____
_____
_____